ISU-GeoBot: A Web-Based Campus Navigation Assistant Integrating an Enhanced RAG Architecture for Faculty Availability Classification

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1. INTRODUCTION

1.1 Background of the Study

University campuses are complex environments where students, faculty, and visitors frequently rely on maps, directories, and physical signage to locate buildings and offices. Traditionally, these navigation tools have been static, offering only geographical information without considering the dynamic nature of campus activities. While such tools help users locate physical structures, they do not provide contextual information such as faculty availability, schedule changes, or real-time academic activity occurring within those locations. This disconnect between spatial navigation and academic operations creates a form of information asymmetry, where the available information does not accurately reflect the real-time status of campus activities.

At the Isabela State University Echague Main Campus, this challenge is particularly noticeable. Current navigation tools mainly consist of printed maps and isolated departmental schedules—individual faculty timetables maintained separately by each department with no centralized or digital access point. These resources are not integrated, making it difficult for users to access contextual information efficiently. Students must often visit multiple locations or consult different offices simply to confirm faculty availability or schedule details. In informal observations conducted by the researchers during the enrollment period, the majority of incoming freshmen reported difficulty in independently locating faculty offices, navigating between college buildings, and determining whether a faculty member was available for consultation. Furthermore, the reliance on unstructured administrative data, such as printed schedules and memoranda, makes it difficult to provide real-time, context-aware information to the campus community.

To address these limitations, modern universities are transitioning toward smart campus environments by integrating spatial databases and intelligent information systems. Emerging artificial intelligence approaches, particularly large language models and Retrieval-Augmented Generation (RAG) systems, have shown strong potential for converting unstructured text into structured, queryable information. However, while standard RAG systems excel at retrieving static facts, they lack the probabilistic capabilities required to estimate dynamic, future states based on historical schedules.

Therefore, this study proposes enhancing the standard Retrieval-Augmented Generation (RAG) architecture by embedding a probabilistic machine learning component—specifically, a Random Forest classifier—directly into the retrieval pipeline. This proposed 'Enhanced RAG' architecture is implemented through ISU-GeoBot, a web-based campus navigation assistant designed to address the challenges of spatial disorientation and dynamic faculty schedules. Unlike standard RAG systems, which only retrieve static documents to ground LLM responses, this system augments its responses with real-time probabilistic analytics derived from temporal schedule data. This allows the system to augment its responses not only with retrieved institutional documents but also with real-time probabilistic analytics derived from temporal schedule data. To adhere to ethical data privacy standards and prevent invasive physical tracking the probabilistic output, undergoes a status masking protocol, transforming raw probability estimates into generalized faculty availability statuses (e.g., “Available for Consultation,” “Currently in a Lecture,” or “Unavailable”) rather than exposing exact physical coordinates.

By integrating this probabilistic modeling with geospatial data and document retrieval, ISU-GeoBot aims to transform traditional campus navigation into a highly responsive information service. Ultimately, the system seeks to reduce spatial disorientation and modernize student services through data-driven technologies, while maintaining strict adherence to faculty privacy and ethical software design.

1.2 Objectives of the Study

General Objective

The primary objective of this study is to develop ISU-GeoBot, a web-based campus navigation assistant that integrates a Random Forest classifier into a Retrieval-Augmented Generation (RAG) architecture to provide context-aware, privacy-compliant probabilistic estimates of faculty availability for the ISU Echague Main Campus.

Specific Objectives

To achieve the general objective, the study will pursue the following specific objectives:

1. To integrate a Random Forest classifier into the Retrieval-Augmented Generation (RAG) pipeline to estimate the probability of real-time faculty availability based on temporal schedule data.
2. To evaluate and compare the performance of the standard and Enhanced RAG architectures in terms of Response Time and RAGAS metrics (Context Precision, Context Recall, Faithfulness, and Answer Relevancy).

3. To deploy the Enhanced RAG architecture within the web-based ISU-GeoBot system to provide context-aware campus navigation and privacy-compliant faculty availability information.
4. To evaluate the functional accuracy and reliability of the system's faculty availability probability estimates through ground-truth validation by selected faculty members.

1.3 Scope and Delimitation

This study focuses on the enhancement of the Retrieval-Augmented Generation (RAG) architecture by integrating a Random Forest classifier to create what the researchers term an “Enhanced RAG” architecture. The enhancement specifically involves embedding a probabilistic machine learning component into the RAG pipeline, enabling the system to augment its responses not only with retrieved institutional documents but also with real-time probabilistic analytics derived from temporal faculty schedule data. This Enhanced RAG architecture is applied within ISU-GeoBot, a web-based intelligent campus navigation assistant designed for the Isabela State University Echague Main Campus. The geospatial coverage of the system will encompass the major academic buildings, colleges, and administrative offices within the ISU Echague Main Campus.

To address ethical considerations regarding continuous location tracking and data privacy, the system will implement status masking. The probabilistic Random Forest model will analyze temporal schedule data to output a generalized faculty availability status (e.g., “Available for Consultation,” “Currently in a Lecture,” or “Unavailable”) rather than outputting exact physical room coordinates.

The application will be implemented as a web-based system to ensure accessibility across different devices. The core contribution of this study is the integration and evaluation of the Enhanced RAG architecture; therefore, the Technical AI Evaluation comparing the standard RAG and the Enhanced RAG using RAGAS metrics (Context Precision, Context Recall, Faithfulness, and Answer Relevancy) will serve as the primary evaluation methodology. A Additionally, 15 selected faculty members and department staff will serve as ground-truth validators to verify the functional accuracy and reliability of the system's faculty availability probability estimates across at least five distinct academic departments.

1.4 Significance of the Study

The results of this study are expected to benefit the following stakeholders:

Students: Students will benefit from improved campus navigation through context-aware location guidance and access to faculty availability information. This can reduce spatial disorientation and improve the efficiency of navigating university facilities.

1. **Faculty Members:** Faculty members may benefit from improved communication with students, as the system can provide estimated availability status based on schedule data, reducing interruptions and unnecessary office visits.
2. **University Administrators:** University administrators may utilize the system as a digital tool for managing and disseminating campus information, supporting institutional efforts to modernize student services and campus management.

3. **The Institution:** The Isabela State University Echague Main Campus may benefit from implementing an intelligent campus navigation system that enhances student services without requiring additional physical infrastructure such as information kiosks.
4. **Future Researchers:** Future researchers may use this study as a reference for further developments in intelligent campus systems, geospatial information systems, and artificial intelligence applications in higher education environments.
5. **The Body of Knowledge:** This study contributes to the existing body of knowledge by demonstrating a novel integration of probabilistic machine learning with Retrieval-Augmented Generation in a localized campus context. The proposed Enhanced RAG architecture, which embeds a Random Forest classifier within the retrieval pipeline, offers a replicable model for future studies seeking to combine probabilistic analytics with generative AI for context-aware information delivery.

2. REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Review of Related Literature

This section reviews the literature and studies relevant to the enhancement of the RAG architecture and implementation of ISU-GeoBot. The discussion is organized thematically, beginning with the evolution of smart campus navigation systems and the integration of artificial intelligence in educational frameworks.

2.1.1 Smart Campus Navigation and Information Systems

The transition from traditional, static campus maps to dynamic, data-driven environments has become a critical focus in modern educational infrastructure. Haggag et al. (2025) note that sustainable smart campus development increasingly relies on integrating advanced technologies to improve operational efficiency. A primary driver for this shift is the persistent issue of spatial disorientation among students and visitors navigating complex university layouts. To address this, Hoang et al. (2025) demonstrated the effectiveness of enabling smart campus indoor spaces through advanced spatial modeling, which provides more granular navigation assistance.

However, navigation is no longer limited to merely finding physical coordinates. Nayakwadi et al. (2025) emphasize that enhancing wayfinding requires the integration of Artificial Intelligence (AI) and the Internet of Things (IoT) to provide real-time, context-aware routing. This aligns with the broader vision of designing the future smart campus, where integrating key elements—such as spatial data and real-time personnel availability—significantly enhances the overall user experience by bridging the gap between geographic location and academic activity (Zhang et al., 2025).

2.1.2 Artificial Intelligence in Educational Frameworks

Beyond spatial navigation, universities are increasingly leveraging AI to manage and disseminate complex administrative data. Kaur et al. (2025) highlight that future education frameworks must integrate IT and business logic to streamline student services. One significant challenge in these environments is the reliance on unstructured textual data, such as university memoranda, schedules, and curriculum guides.

To combat information overload, educational institutions have begun deploying AI-driven systems to convert this unstructured data into queryable formats. For example, Swacha and Gracel (2025) surveyed the application of AI chatbots in education, noting their growing utility in providing instant academic support. Similarly, Mazzei et al. (2025) developed an AI system designed to provide high school students with accurate academic guidance, demonstrating how language models can effectively process institutional data to support student decision-making.

2.1.3 Probabilistic Classification using Random Forest

In dynamic campus environments, estimating the probability of the status or availability of resources requires robust machine learning algorithms capable of handling temporal data. The Random Forest algorithm, an ensemble learning method, has proven highly effective for classification tasks involving schedule-based probability . Alam et al. (2024) successfully applied Random Forest models with timestamp features to detect room occupancy, showcasing the algorithm's ability to identify patterns in temporal data and accurately estimate real-time status

This probabilistic capability is essential for modern campus systems that aim to estimate the actual availability of faculty members—a task complicated by the inherent stochasticity of human behavior. While static schedules provide a deterministic baseline, real-world faculty presence is influenced by numerous dynamic factors including meeting overruns, university-wide events, personal habits, and seasonal academic patterns. By processing these multidimensional temporal and behavioral features through a Random Forest classifier, systems can output probabilistic estimates regarding whether a specific faculty member is currently available, going beyond what simple rule-based schedule lookups can achieve.

2.1.4 Retrieval-Augmented Generation (RAG) Architecture

While traditional Large Language Models (LLMs) are powerful conversational agents, they are prone to 'hallucinations' and lack access to private, real-time institutional data. Retrieval-Augmented Generation (RAG) addresses these limitations by forcing the model to retrieve relevant information from an external database before generating a response (Brown, Roman, & Devereux, 2025). This architecture ensures that responses are grounded in accurate, domain-specific documents.

The effectiveness of RAG has been validated across multiple domains. Lang and Gürpinar (2025) confirmed the efficacy of RAG-powered chatbots in providing learning support within online courses, ensuring students receive answers strictly based on course materials. Furthermore, Mahdhin et al. (2025) emphasized that RAG significantly enhances the precision and interpretability of LLMs when handling highly specific textual data, making it an ideal architecture for processing university schedules and department announcements.

2.1.5 Data Privacy and Ethical Tracking in Campuses

As smart campuses become more interconnected, the ethical implications of data collection and location tracking have emerged as significant concerns. Cheong and Nyaupane (2022) investigated the growing student tensions surrounding 'dataveillance' and the governance of IoT data in universities, warning that overly invasive tracking can foster distrust among the campus community.

While location-aware technologies provide undeniable utility, they introduce critical challenges related to privacy and the handling of sensitive location data (Kanapathy et al., 2026). Consequently, ethical system design mandates that applications balance functional utility with robust privacy safeguards. Implementing strategies such as 'status masking'—where a system outputs a generalized availability status rather than exact physical coordinates—serves as a necessary compromise to protect faculty privacy while still delivering context-aware navigation assistance.

2.1.6 Synthesis of the State-of-the-Art

The review of related literature highlights significant advancements in isolated technologies: smart campus mapping (Hoang et al., 2025), RAG-based document retrieval (Brown, Roman, & Devereux, 2025), and probabilistic modeling for occupancy (Alam et al., 2024). Current research demonstrates that while traditional LLMs struggle with hallucinations, RAG architectures successfully ground AI responses in verified local data. Furthermore, machine learning classifiers like Random Forest are highly proven in processing temporal data for real-time status probability estimates. However, all existing RAG implementations in the reviewed literature operate as purely retrieval-based systems—they retrieve static documents but do not incorporate predictive analytics into the retrieval pipeline. This distinction is critical, as it defines the boundary between standard RAG and the Enhanced RAG architecture proposed in this study.

2.1.7 Gap of the Study

Despite these individual advancements, a distinct research gap exists in the integration of these distinct methodologies into a single, cohesive campus assistant tailored for localized academic environments. Current smart campus systems generally lack the capability to fuse probabilistic classification(like faculty schedule logic) with generative text retrieval within a unified pipeline. Specifically, no existing study has demonstrated an Enhanced RAG architecture—one that augments the standard document-retrieval process with real-time probabilistic outputs from a machine learning classifier. Furthermore, there is a notable absence of literature demonstrating how to safely implement these probabilistic location systems without violating faculty privacy through invasive GPS tracking. This study addresses both gaps by proposing an Enhanced RAG pipeline that integrates Random Forest probability estimates with document retrieval, mediated by a privacy-preserving status masking protocol.

2.1.8 Relevance of the Reviewed Literature to the Present Study

The reviewed literature provides the theoretical and technical foundation necessary to construct ISU-GeoBot and fill the identified research gap. The studies on Random Forest (Alam et al., 2024) directly inform the system's probabilistic classification logic. The literature on RAG architectures (Mahdhin et al., 2025) guides the development of the system's document retrieval pipeline to prevent AI hallucinations. Finally, the ethical frameworks regarding campus "dataveillance" (Cheong & Nyaupane, 2022) justify the implementation of the system's "Status Masking" protocol. Together, these studies validate the technical feasibility and ethical necessity of the proposed hybrid architecture.

2.2 Theoretical and Conceptual Framework

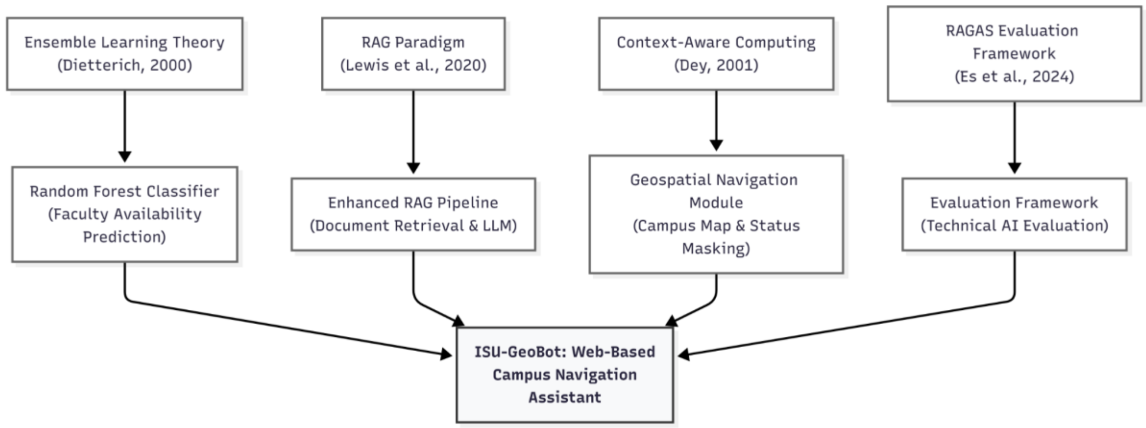

Figure 1. Theoretical Framework of ISU-GeoBot

The enhancement of the RAG architecture within ISU-GeoBot is anchored on the following theoretical foundations that inform its core architectural components:

Ensemble Learning Theory. Ensemble learning is a machine learning paradigm where multiple models are combined to produce a more robust and accurate estimate than any individual model could achieve (Dietterich, 2000). The Random Forest algorithm, a prominent ensemble method, constructs multiple decision trees and aggregates their outputs through majority voting for classification tasks. This theory provides the foundational justification for employing Random Forest as the probabilistic engine for faculty availability classification in the ISU-GeoBot system, where temporal schedule patterns must be accurately interpreted to produce reliable status probability estimates.

Retrieval-Augmented Generation (RAG) Paradigm. The RAG paradigm, formalized by Lewis et al. (2020), addresses the inherent limitations of standalone Large Language Models (LLMs) by coupling a parametric language model with a non-parametric retrieval mechanism. This hybrid approach allows the model to access and incorporate domain-specific external knowledge during response generation, significantly reducing hallucinations and improving factual accuracy. This theoretical basis directly informs the design of ISU-GeoBot's document retrieval pipeline, where the LLM is constrained to generate responses grounded in verified institutional documents rather than relying solely on its pre-trained parameters.

Context-Aware Computing. Context-aware computing, as defined by Dey (2001), refers to systems that use contextual information—such as location, time, identity, and activity—to provide relevant information and services to users. This theory underpins the fundamental design philosophy of ISU-GeoBot, which goes beyond static map-based navigation by dynamically adapting its responses based on temporal context (time of day, day of the week) and personnel-related context (faculty schedules and availability).

RAGAS Evaluation Framework. The RAGAS (Retrieval-Augmented Generation Assessment) framework, proposed by Es et al. (2024), provides a standardized set of automated metrics specifically designed for evaluating the quality of RAG pipelines. Its four core metrics—Context Precision, Context Recall, Faithfulness, and Answer Relevancy—enable objective, reproducible comparison between different RAG architectures. This framework provides the theoretical basis for the study's primary evaluation methodology, ensuring that the assessment of the Enhanced RAG architecture is grounded in domain-specific, automated metrics rather than subjective judgment.

Based on these theoretical foundations, the conceptual framework of this study operates on an Input-Process-Output (IPO) model, which is divided into three distinct operational phases to illustrate how the theories are operationalized within the ISU-GeoBot system architecture:

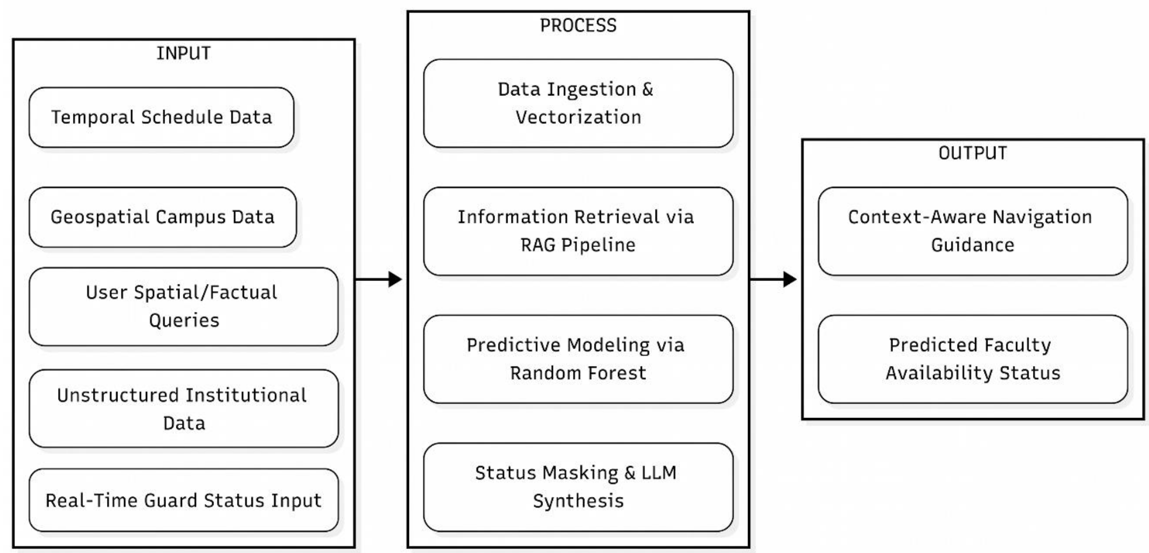

Figure 2. The Input-Process-Output (IPO) Model of ISU-GeoBot

Phase 1: Input Stage. This foundational phase involves gathering the necessary raw data for the system to function. It ingests natural language user queries alongside dynamic temporal schedule data, static geospatial campus coordinates, unstructured institutional documents (such as university memoranda), and real-time manual campus presence logs provided by security personnel.

Phase 2: Hybrid AI Process. This phase represents the core logical architecture of the system, operationalized through the proposed Enhanced RAG architecture. The Random Forest algorithm analyzes the temporal schedules to classify faculty availability, which is immediately subjected to a status masking protocol to ensure ethical data privacy. Simultaneously, the Retrieval-Augmented Generation (RAG) pipeline processes the unstructured institutional and geospatial data. A Large Language Model (LLM) then acts as the central synthesizer, merging the masked probabilistic status with the retrieved spatial context.

Phase 3: Output Stage. In the final phase, the system delivers the synthesized information back to the user. This results in a generalized, privacy-compliant estimated faculty availability status, coupled with context-aware navigation guidance tailored specifically to the Isabela State University Echague Main Campus environment.

2.3 Definition of Terms

To ensure clarity and precision, the following terms are defined based on how they are operationalized within the context of this study:

1. Context-Aware Navigation - An intelligent navigation approach that goes beyond static maps by utilizing contextual information such as real-time events or personnel schedules to provide adaptive, user-friendly, and personalized route recommendations.
2. Geospatial Data - Information that represents objects, events, or features that have a specific physical location on the surface of the Earth. This data is often large and requires specialized databases for indexing and querying.
3. Hallucination (AI) - A phenomenon in which a Large Language Model generates responses that appear fluent and confident but contain fabricated, inaccurate, or unverifiable information. In this study, hallucination is the primary limitation that the Enhanced RAG architecture is designed to mitigate by grounding all LLM responses in retrieved institutional documents and verified predictive outputs.

4. Random Forest - A supervised machine learning algorithm that constructs an ensemble of multiple decision trees to reach a single, more precise probabilistic result.

5. Retrieval-Augmented Generation (RAG) - A technique that enhances large language models by retrieving and incorporating relevant information from external data sources before generating responses. This allows the system to use domain-specific and updated information, such as campus documents, without needing to undergo costly retraining.

6. Smart Campus - A dynamic educational environment that integrates physical infrastructure with digital technologies, big data, and artificial intelligence to optimize operations, resource management, and the overall user experience.

7. Spatial Disorientation - The difficulty users experience when trying to maintain their orientation or navigate through complex environments, a challenge that is often amplified in large building layouts.

8. Unstructured Data - Information that does not conform to a predefined data model or reside in a traditional row-column database. In the context of this study, this includes text-heavy administrative documents, PDFs, and image files.

9. Status Masking - A privacy-preserving protocol in which a system transforms raw probabilistic outputs into generalized, non-identifying status categories before presenting them to the user. This

technique ensures that sensitive data, such as the exact physical location of personnel, is never exposed.

10. Enhanced RAG - A novel architecture proposed in this study that extends the standard Retrieval-Augmented Generation framework by integrating a probabilistic machine learning component (Random Forest classifier) into the retrieval pipeline. This allows the system to generate responses grounded in both retrieved documents and real-time probabilistic analytics.

11. Large Language Model (LLM) - A type of artificial intelligence model trained on vast amounts of text data, capable of understanding and generating human-like natural language. In this study, the LLM serves as the central response synthesizer that merges retrieved documents and predictive outputs into coherent, conversational answers.

12. Temporal Schedule Data - Time-dependent information derived from faculty class schedules, including features such as the day of the week, time of day, and semester period. In this study, temporal schedule data serves as the primary input for the Random Forest classifier to estimate faculty availability status at any given point in time.

13. Vector Embeddings - Numerical vector representations of text data generated by a machine learning model. These fixed-length arrays of numbers capture the semantic meaning of text, enabling mathematical operations such as cosine similarity to measure how closely related two pieces of text are. In this study, vector embeddings are used to convert

institutional documents and user queries into a comparable numerical format for semantic retrieval.

14. Functional Validation - A systematic evaluation process in which domain experts interact with a completed system to verify whether its outputs are factually accurate and meet the specified functional requirements. In this study, functional validation involves faculty members using ISU-GeoBot and confirming whether the system's estimated availability statuses match their actual real-world presence.

15. Semantic Similarity - A computational measure of how closely related two pieces of text are in meaning, rather than in exact wording. In this study, semantic similarity is used to match user queries against stored document embeddings to retrieve the most contextually relevant information from the knowledge base.

16. Microservice Architecture - A software design pattern in which an application is composed of small, independently deployable services that communicate over a network. In this study, the Random Forest classification model is deployed as a separate Python-based microservice that the main Node.js backend invokes via HTTP requests.

17. Cosine Similarity - A mathematical metric used to measure the angular distance between two vectors in a multi-dimensional space, yielding a value between 0 (no similarity) and 1 (identical). In this study, cosine similarity is the specific function used to rank the relevance of stored

document embeddings against a user query embedding during the retrieval phase of the RAG pipeline.

18. Context Fusion **-** The core architectural mechanism within the Enhanced RAG pipeline that enables the dynamic integration of real-time predictive analytics, such as masked faculty availability status, with retrieved institutional document chunks. This process facilitates the synthesis of multiple, distinct data streams into a unified, comprehensive prompt prior to being processed by the Large Language Model for final response generation.

3. METHODOLOGY

3.1 Research Design

This study will employ a Developmental Research Design, which focuses on the systematic design, development, and evaluation of a technological artifact intended to solve an identified problem (Richey & Klein, 2014). This research design is appropriate for the study because the primary goal is to enhance the RAG architecture and implement it within ISU-GeoBot, then evaluate its quality against established software standards rather than to test the causal relationship between variables.

The evaluation component of this study will focus on two complementary analyses. First, a Technical AI Evaluation will be conducted using the RAGAS (Retrieval-Augmented Generation Assessment) framework (Es et al., 2024) to scientifically compare the response quality of the standard RAG architecture against the proposed Enhanced RAG architecture using established metrics: Context Precision, Context Recall, Faithfulness, and Answer Relevancy. Second, a Functional Validation will be performed by 15 selected faculty members and department staff, who will serve as ground-truth validators to verify the accuracy and reliability of the system's faculty availability probability estimates against their actual presence. The resulting data from both evaluations will be analyzed using their respective analytical methods as described in Section 3.9.

3.2 Research Locale

The study will be conducted at the Isabela State University (ISU) Echague Main Campus, located in Echague, Isabela, Philippines. The campus serves as the main hub of academic and administrative operations for the university and comprises multiple buildings, colleges, and administrative offices distributed across a wide geographic area.

ISU Echague Main Campus was selected as the research locale because it is a large, multi-building campus environment where the problem of spatial disorientation is particularly relevant. Students and visitors frequently encounter difficulties in navigating the campus and locating faculty members due to the absence of an integrated digital navigation and information system.

3.3 Respondents of the Study

The respondents of this study will consist of 15 selected faculty members and department staff of the Isabela State University Echague Main Campus. Faculty members and department staff were selected because they are the only individuals who possess the ground-truth knowledge necessary to validate the accuracy of the system's faculty availability estimates. Unlike general end-users, faculty members can directly confirm or deny whether the system's predicted availability status (e.g., "Available for Consultation," "Currently in a Lecture," or "Unavailable") corresponds to their actual real-world presence at any given time. This makes them the most qualified respondent group for evaluating the Functional Accuracy and Reliability of the Enhanced RAG architecture's probabilistic component.

Purposive sampling will be employed to select the faculty respondents. Purposive sampling is a non-probability technique in which respondents are deliberately selected based on predefined criteria relevant to the research objectives (Creswell & Creswell, 2018). The criteria for selection are: (1) the faculty member's schedule data is included in the system's training dataset, (2) they are currently active and teaching at ISU Echague Main Campus, and (3) they are willing to use the system and verify the accuracy of its availability estimates. A sample of 15 faculty members is deemed sufficient for this validation phase, as they serve as expert validators and ground-truth providers rather than a general population sample from which statistical generalizations are drawn. Nielsen (1994) demonstrated that 5 expert evaluators can identify approximately 85% of system issues, while Faulkner (2003) showed that 15 evaluators achieve approximately 97% problem detection coverage. The selection will target representation across at least five distinct academic departments to ensure that the system's probabilistic estimates are tested against varied schedule patterns, teaching loads, and attendance behaviors.

3.4 Research Procedures

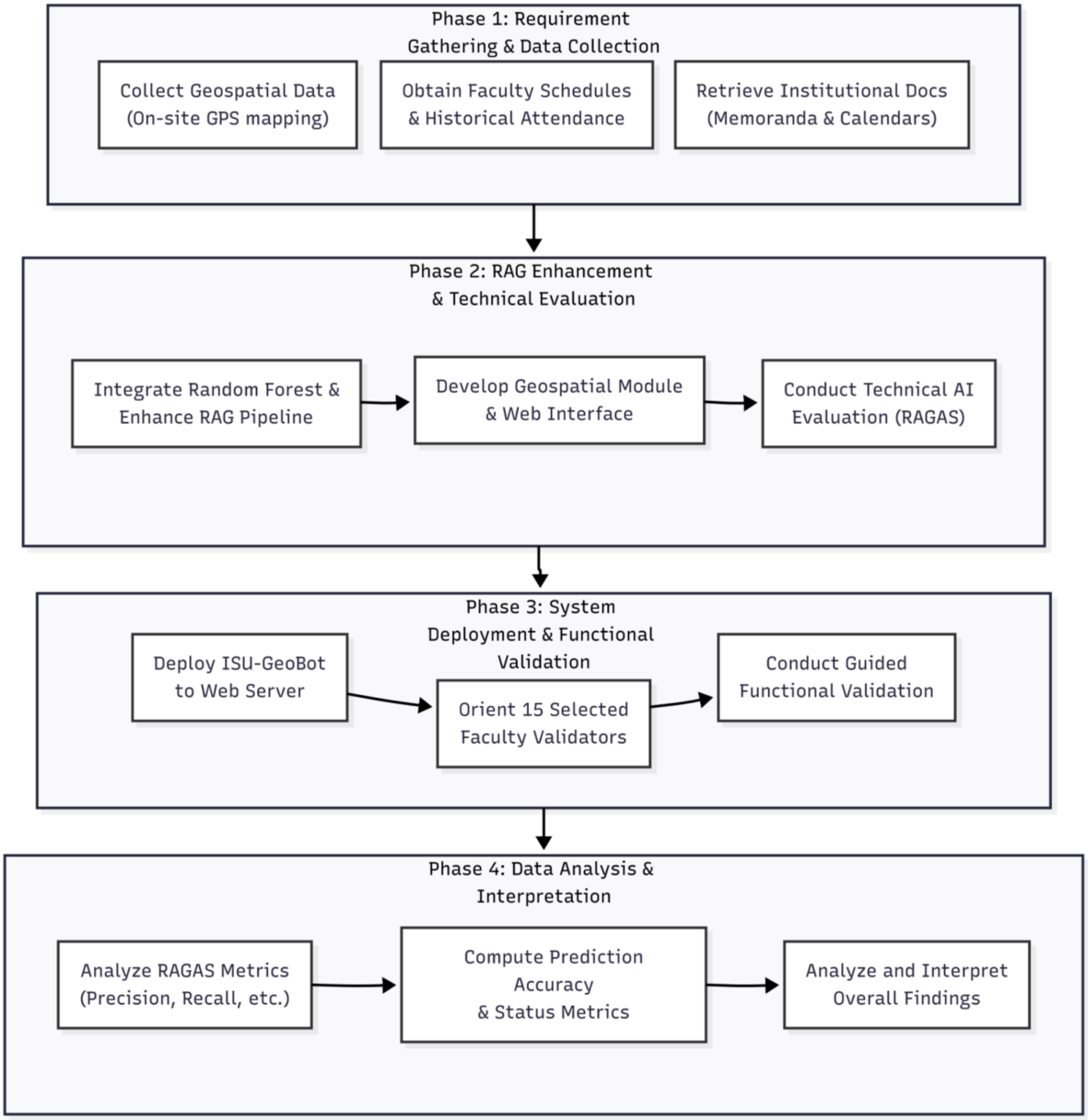

Figure 3. Research Procedures Flowchart

The research process for this study will be carried out in four sequential phases:

Phase 1: Requirement Gathering and Data Collection. The researchers will collect the necessary raw data required for the system to function. This will include (a) geospatial data such as campus building coordinates and points of interest, collected through on-site GPS mapping and verification with campus administrators; (b) faculty schedule data, obtained with permission from the respective department heads and the university registrar; (c) institutional documents such as university memoranda, announcements, and academic calendars, retrieved from official university channels; and (d) historical faculty attendance records, obtained from departmental logbooks or HR biometric attendance systems. To maintain ethical data privacy standards, all attendance records will be sanitized and anonymized—removing personally identifiable information and retaining only temporal patterns (e.g., check-in times, frequency of presence during scheduled hours)—before being processed as training features for the Random Forest classifier.

Phase 2: RAG Enhancement, Integration, and Technical Evaluation. Using the collected data, the researchers will enhance the standard Retrieval-Augmented Generation (RAG) architecture by integrating the Random Forest classifier into the retrieval pipeline. This phase involves training the Random Forest model on the collected faculty schedule data and historical attendance records, implementing the status masking protocol, and embedding the probabilistic output as additional context within the RAG pipeline. Once both the standard RAG and Enhanced RAG pipelines are operational, the researchers will conduct the Technical AI Evaluation. A curated test dataset of representative campus queries (e.g., faculty availability, building directions, schedule inquiries) will be prepared, and their corresponding ground-truth answers will be manually composed by the researchers and validated by department staff to serve as the objective benchmark for the RAGAS evaluation. Each query will be processed by both the standard RAG and the Enhanced RAG, and the outputs will be evaluated against the ground-truth answers using the RAGAS framework metrics: Context Precision, Context Recall, Faithfulness, and Answer Relevancy. This controlled comparison will provide objective, quantitative evidence of the Enhanced RAG's improvement over the baseline. The remaining system components, including the geospatial navigation module and web interface, will be developed following the methodology described in Section 3.6.

Phase 3: System Deployment and Functional Validation. Upon completion of development, internal testing, and the Technical AI Evaluation (Phase 2), the system will be deployed on a web server. The 15 selected faculty members and department staff will be given a structured orientation on how to use the system. During a guided evaluation period, they will interact with ISU-GeoBot by querying their own availability status and verifying whether the system's probability estimates match their actual presence. Each faculty validator will complete a structured validation checklist recording the accuracy of the system's probability estimates across multiple time points and scenarios.

Phase 4: Data Analysis and Interpretation. The results from the Technical AI Evaluation (Phase 2) and the Faculty Functional Validation (Phase 3) will be compiled and analyzed. The RAGAS metrics will be reported as comparative scores between the standard RAG and Enhanced RAG pipelines. The faculty validation data will be analyzed to compute the classification accuracy rate(percentage of correct classifications out of total classifications verified) and to identify patterns in classification errors across departments, time slots, and event periods. The combined results will provide both automated and human-verified evidence of the Enhanced RAG architecture's effectiveness.

3.4.1 Data Gathering

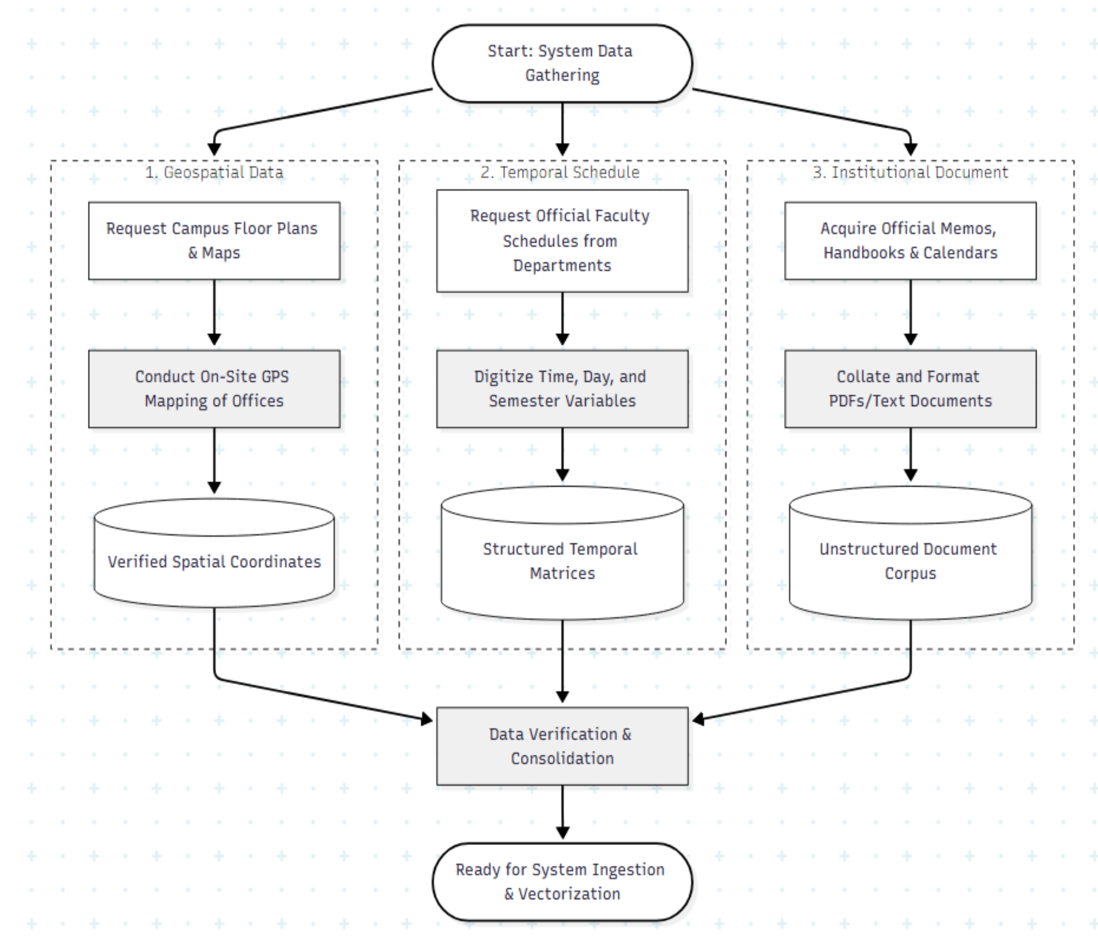

Figure 4. Data Gathering Diagram

The data collection for this study follows three parallel streams, as illustrated in Figure 4. Each stream targets a specific category of data required by the ISU-GeoBot system's core modules.

(a) Geospatial Data Collection. The researchers will request campus floor plans and building maps from the university administration to establish a baseline spatial reference. On-site GPS mapping will then be conducted using mobile devices to record the geographic coordinates (latitude and longitude) of all buildings, offices, departments, and key points of interest within the ISU Echague Main Campus. The collected GPS coordinates will be verified and cross-referenced against physical landmarks to ensure spatial accuracy. The output of this process is a verified spatial coordinates dataset that will serve as the foundation for the geospatial navigation module described in Section 3.5.1.

(b) Temporal Schedule and Attendance Collection. Official faculty class schedules will be requested from department heads across the participating colleges. These schedules will be digitized into structured temporal variables, including day of the week, time slot, semester period, and teaching load. Additionally, historical attendance logs will be obtained from departmental logbooks or HR biometric records. These records will be sanitized and anonymized to remove personally identifiable information in compliance with the Data Privacy Act of 2012 (RA 10173). The resulting output is a structured temporal and attendance feature matrix that will be used to train the Random Forest prediction module described in Section 3.5.2.

(c) Institutional Document Collection. Official university memoranda, student handbooks, academic calendars, and other relevant institutional documents will be acquired from administrative offices. These documents will be collated and formatted into machine-readable PDF or plain text files. Each document will then be segmented into smaller text chunks suitable for vector embedding and semantic retrieval. The output is an unstructured document corpus that will be ingested into the RAG pipeline described in Section 3.5.4.

Upon completion of all three streams, a data verification and consolidation step will ensure completeness and consistency across all datasets before they are ingested into the system for vectorization and model training.

3.5 System Architecture

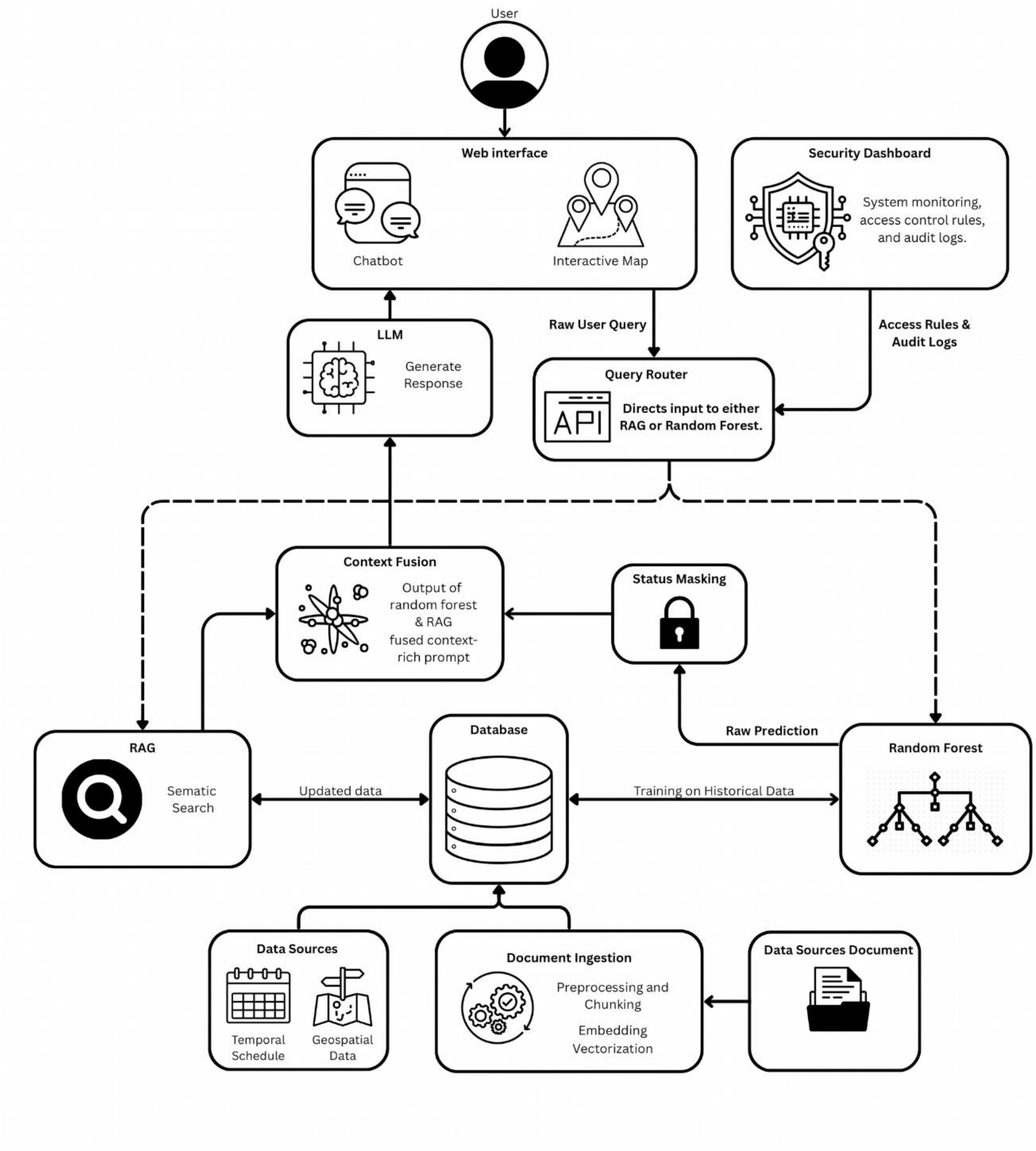

Figure 5. ISU-GeoBot System Architecture Diagram

This section describes the technical architecture of ISU-GeoBot, detailing the design and function of each core module. The system follows a three-tier architecture consisting of a web-based frontend client, a backend application server housing the AI processing engine, and a database layer for storing geospatial, schedule, and document data. The overall architecture is composed of four integrated modules that operate in coordination to process user queries and deliver context-aware responses. The frontend module provides the user interface, including an interactive campus map, a chatbot interface, and a dedicated security dashboard for manual campus presence tracking. User queries are transmitted to the backend server, which routes requests to the appropriate processing module: the Random Forest Prediction Module for faculty availability queries, or the RAG Pipeline for document-based information retrieval. The outputs from these modules are synthesized by the Large Language Model (LLM) and returned to the user as a natural language response.

3.5.1 Geospatial Navigation Module

The geospatial navigation module will serve as the spatial backbone of the system. Campus building coordinates, office locations, and key points of interest will be stored in a geospatial database. The frontend will render an interactive campus map using a JavaScript mapping library, enabling users to visually locate buildings, offices, and navigate between campus locations. Each geospatial data point will be associated with contextual metadata, such as the department name, building function, and linked faculty information, enabling the system to provide location-aware responses to user queries.

3.5.2 Random Forest Classification Module

The random forest classification module will be responsible for classifying faculty availability based on a combination of temporal schedule data and historical behavioral patterns. Unlike a simple rule-based schedule lookup (e.g., SQL IF/ELSE queries), which can only confirm whether a faculty member has a class at a given time, the Random Forest model is designed to estimate the probability that a faculty member is actually present and available—a distinction that accounts for real-world stochastic variability. The module will process the following categories of input features: (a) static temporal features, including day of the week, time of day, and semester period derived from official faculty schedule entries; (b) historical attendance patterns, including aggregated records of past faculty check-ins or sign-ins that capture individual tendencies toward punctuality, early departure, or extended office hours; (c) institutional event indicators, such as binary flags denoting university-wide events (e.g., convocations, faculty assemblies, enrollment periods) during which normal schedules are frequently disrupted; and (d) contextual features, such as proximity to examination periods or semester breaks, which historically correlate with changes in faculty presence. These features will be engineered from the raw schedule and historical data to create a multidimensional feature space that captures the inherent uncertainty in human behavior—a task that requires probabilistic machine learning rather than deterministic rule-based logic.

To construct the decision trees within the Random Forest ensemble, the algorithm requires a mathematical metric to evaluate the optimal way to split the data at each node. This study utilizes the Gini Impurity index as the primary splitting criterion. Its role is to measure the probability of incorrectly classifying a randomly chosen faculty status if it were labeled randomly according to the distribution in the subset. By iteratively selecting the feature (e.g., time of day vs. historical attendance) that minimizes the Gini Impurity, the model maximizes the certainty of its classifications to determine the final estimated status. The calculation is defined as:

$$Gini(t) = 1 - \sum_{i=1}^{C} p(i|t)^2$$

Where:

p(i|t) = proportion of samples belonging to class i at node t

C = total number of classes (Available, Late, Absent)

The training process will involve splitting the prepared dataset into training and testing subsets using an 80-20 ratio, and cross-validation will be applied to optimize model performance. Standard classification metrics—including accuracy, precision, recall, and F1-score—will be used to evaluate the model's classification performance. Feature importance analysis will be conducted to identify which categories of input features (e.g., historical attendance vs. static schedule) contribute most to classification accuracy, thereby validating the necessity of the machine learning approach over a simple rule-based alternative. The trained model will output one of the following generalized availability statuses for each faculty member: "Available for Consultation," "Currently in a Lecture," or "Unavailable."

3.5.3 Status Masking Protocol

To ensure compliance with ethical data privacy standards and prevent invasive physical tracking of faculty members, the system implements a strict programmatic status masking protocol. Rather than outputting exact physical coordinates or specific room numbers, the

Random Forest model's raw probabilistic output undergoes a middleware transformation before being passed to the generative AI pipeline.

The masking protocol operates as a three-step data sanitization process:

1. **Raw Inference Capture:** The Random Forest classification module generates a raw classification output, which typically represents a specific physical location class (e.g., Class_Room_302 or Off_Campus_Tag).

2. **Associative Mapping (Hash Map):** The backend server utilizes a predefined associative array (hash map) to intercept this raw output. The hash map links specific, sensitive location keys to a standardized set of generalized status values. For example, if the model outputs a classroom location, the hash map translates this to the string "Currently in a Lecture". If the output is an office location, it translates to "Available for Consultation".

3. **Data Purging and LLM Injection:** Once the translation is complete, the original location variable is purged from the active session memory. Only the sanitized, generalized status string is transmitted as context to the Large Language Model for response synthesis.

At no point will the system store, transmit, or display the exact physical location of any faculty member to the end user. This middleware protocol ensures that the system delivers meaningful availability information while maintaining a hard boundary for data privacy, as discussed in the related literature (Cheong & Nyaupane, 2022).

To illustrate the programmatic execution of this protocol, the following Node.js logic demonstrates the interception, translation, and purging process before the context is injected into the Llama 3.1 8B prompt:

```
// 1. Intercept: Check deterministic guard input from Supabase firstlet

isFacultyOnCampus = await checkGuardLogs(facultyId);

let safeStatus = "";
if (!isFacultyOnCampus) {
    // Deterministic Override: Bypass AI if guard confirmed they left
    safeStatus = "Unavailable";
} else {

// 2. Probabilistic Classification: Let Random Forest estimate status
    let rawPrediction = await fetchRandomForestPrediction(facultyId);

    const statusHashMap = {
        "Room_304": "Currently in a Lecture",
        "Faculty_Lounge": "Available for Consultation"
    };
    safeStatus = statusHashMap[rawPrediction];
    rawPrediction = null; // Purge exact location from memory
}

// 3. Hand-off: Inject only the safe status into the LLM context
let llmPrompt = `Context: The faculty member is ${safeStatus}. User Query: ...`;
```

3.5.4 Enhanced RAG Pipeline and Context Fusion

The Retrieval-Augmented Generation (RAG) pipeline will serve as the intelligent information retrieval engine of the system. Institutional documents—including university memoranda, academic calendars, department announcements, and faculty directory information—will be ingested into the pipeline through the following process:

First, each document will be preprocessed and segmented into smaller text chunks to facilitate efficient retrieval. Second, each chunk will be converted into a numerical vector representation (embedding) using the all-MiniLM-L6-v2 sentence transformer model. Instead of using a standalone vector database, these embeddings will be stored directly within the Supabase PostgreSQL database utilizing the pgvector extension. Third, when a user submits a query, the system will compute the semantic similarity between the query and the stored document embeddings to retrieve the most relevant text chunks.

To retrieve the most relevant information from the database, the system must mathematically determine how closely a user's query matches the stored documents. This study utilizes the Cosine Similarity equation as the primary retrieval metric. Its role is to calculate the semantic similarity between the user's natural language query and the institutional document chunks by measuring the cosine of the angle between their respective vector representations. A score closer to 1 indicates high semantic relevance, allowing the system to accurately rank and fetch the best contextual data to feed into the Large Language Model. The metric is defined as:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

Where:

A = vector embedding of user query

B = vector embedding of stored document chunk

$A \cdot B$ = dot product of vectors A and B

$\|A\|$ = Euclidean norm (magnitude) of vector A

$\|B\|$ = Euclidean norm (magnitude) of vector B

The Enhancement: Context Fusion

The key architectural contribution of this study lies in the Context Fusion mechanism that distinguishes the Enhanced RAG from a standard RAG pipeline. In a conventional RAG architecture, the retrieval component only fetches static document chunks, and the language model generates responses based solely on this retrieved textual context. The Enhanced RAG, however, introduces a real-time probabilistic signal into the generation process by injecting the Random Forest classifier's output as additional context before response generation.

Specifically, when the system identifies a faculty-related query (e.g., "Is Prof. Santos available?"), the Random Forest Classification Module (Section 3.5.2) generates a real-time availability probability estimate based on the structured temporal and attendance features. This probabilistic estimation (e.g., "Available," "In a Lecture," or "Absent") is then processed through the Status Masking Protocol (Section 3.5.3) to ensure that only the faculty member's professional availability status is disclosed without revealing precise location data or movement patterns.

The masked estimate is then fused with the retrieved document chunks into a single, structured prompt. This Context Fusion step combines three distinct information sources into one coherent input for the language model: (1) the user's original natural language query; (2) the semantically relevant document chunks retrieved from the vector store; and (3) the real-time, masked faculty availability status from the Random Forest module. This enriched, multi-source prompt is then sent to the Llama 3.1 8B model (accessed via the Groq API) for final response generation.

This architecture ensures that all generated responses are informed by both verified institutional documents and dynamic, real-time probabilistic analytics, thereby addressing the hallucination problem commonly associated with standalone LLMs (Brown, Roman, & Devereux, 2025; Lewis et al., 2020) while simultaneously providing dynamic, context-aware information that standard RAG systems cannot deliver.

Enhanced RAG Process Flow

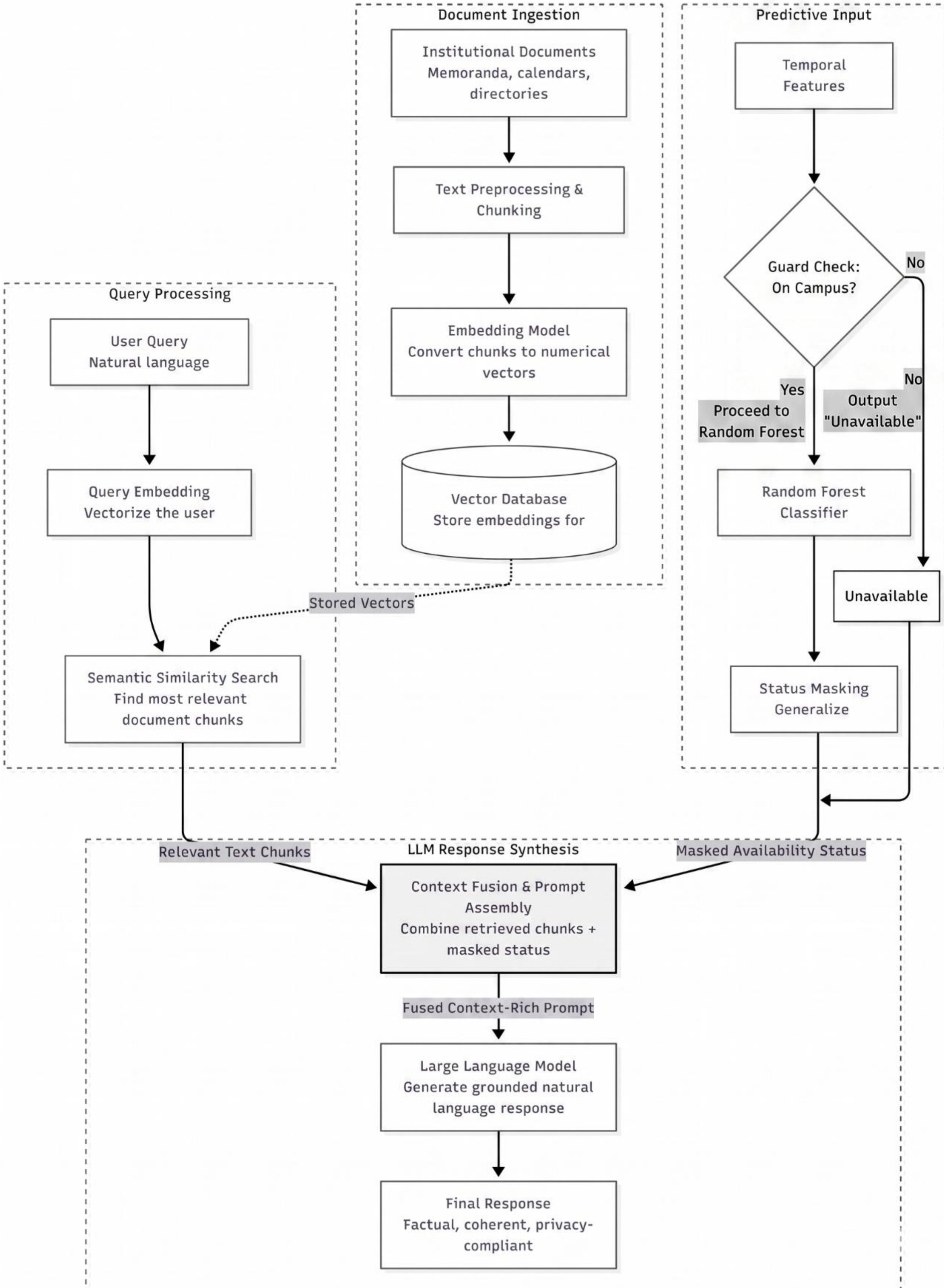

Figure 6. Enhanced RAG Pipeline Architecture

The Enhanced RAG process operates as follows: (1) the user submits a natural language query through the chatbot interface; (2) the system determines whether the query involves faculty availability, campus navigation, or general institutional information; (3) or faculty-related queries, the Status Masking Protocol first checks the deterministic security logs; if the faculty is on-campus, the Random Forest module generates an estimated availability status, which is then passed through the protocol to produce a privacy-compliant output; (4) simultaneously, the RAG pipeline retrieves the most semantically relevant document chunks from the vector store based on the query embedding; (5) a structured prompt is constructed that combines the user's original query, the retrieved document context, and the masked faculty status into a single input; and (6) this combined prompt is sent to the Llama 3.1 8B model, which synthesizes all inputs into a coherent, grounded, natural language response.

Figure 6 illustrates the complete Enhanced RAG Pipeline Architecture, showing the flow from user query to the Context Fusion mechanism and final response generation.

3.6 System Development Methodology

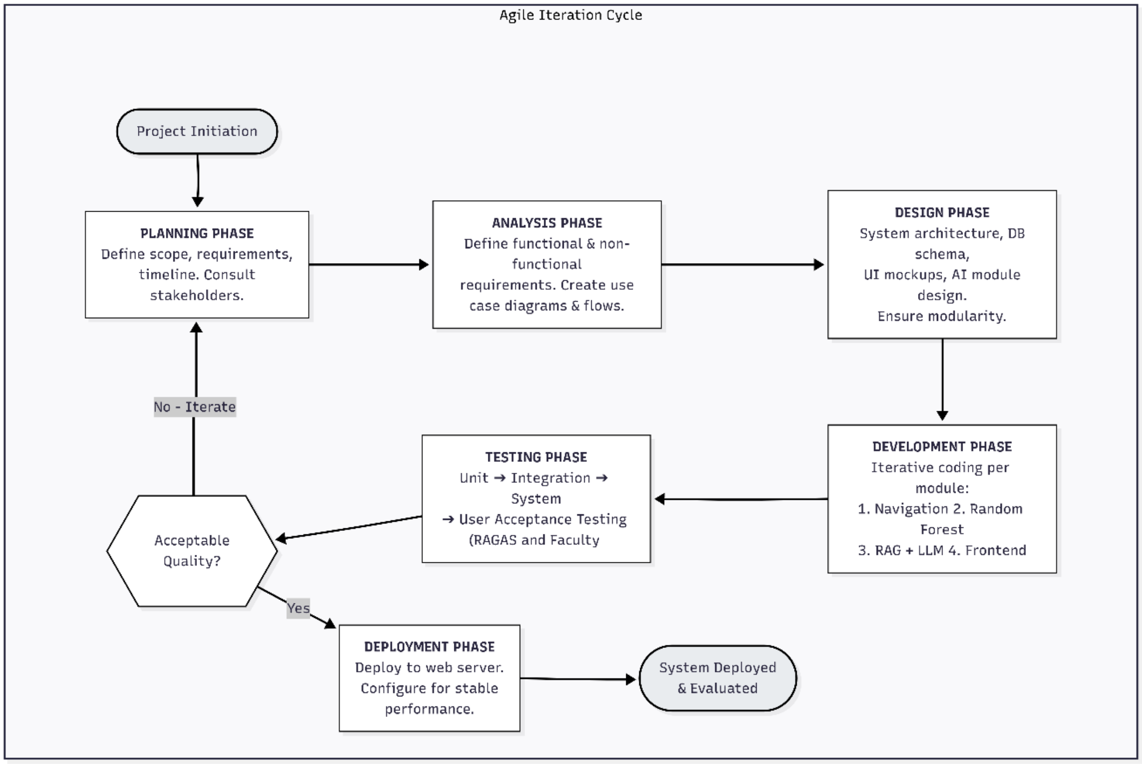

Figure 7. Agile Development Methodology Cycle

The development of ISU-GeoBot will follow the Agile Development Methodology, specifically utilizing an iterative and incremental approach. This methodology was chosen because it allows for continuous refinement of system features through repeated cycles of planning, development, testing, and feedback (Sommerville, 2016). Given the complexity of integrating multiple AI components (Random Forest, RAG, and LLM), an iterative approach provides the flexibility to address technical challenges as they arise without disrupting the overall development timeline.

The development process will be organized into the following phases:

3.6.1 Planning Phase

This phase will involve defining the project scope, identifying system requirements, and establishing the development timeline. The researchers will consult with university stakeholders, including department heads and the campus administration, to confirm the requirements for campus data, faculty schedule access, and institutional document retrieval.

3.6.2 Analysis Phase

During this phase, the researchers will analyze the requirements gathered during the planning phase to define the functional and non-functional requirements of the system. Use case diagrams and system process flows will be developed to model the expected user interactions and system behaviors.

3.6.3 Design Phase

The design phase will produce the system's technical blueprints, including the overall system architecture diagram, database schema design (entity-relationship diagram), user interface mockups, and the detailed design of each AI module. The design will ensure that all components are modular and can be developed and tested independently.

3.6.4 Development Phase

The actual coding and implementation of the system will take place during this phase. Development will proceed iteratively, with each iteration focusing on a specific module: (1) the geospatial navigation module, (2) the Random Forest classification module and status masking protocol, (3) the RAG pipeline and LLM integration, and (4) the web-based frontend interface. Each module will undergo unit testing upon completion before integration with other modules.

3.6.5 Testing Phase

System testing will be conducted at multiple levels. Unit testing will verify the correctness of individual modules. Integration testing will ensure that the modules function correctly when combined. System testing will validate the overall functionality of ISU-GeoBot against the defined requirements. The Technical AI Evaluation using the RAGAS framework will be conducted to compare the standard RAG against the Enhanced RAG pipeline. Finally, the Faculty Functional Validation will be conducted with the 15 selected faculty validators, who will verify the accuracy of the system's probability estimates using structured validation checklists.

3.6.6 Deployment Phase

Upon successful completion of testing, the system will be deployed on a web server accessible to the respondents for the evaluation period. The deployment will be configured to ensure stable performance during the evaluation period.

3.7 Tools and Technologies Used

The following tools and technologies will be utilized in the implementation of ISU-GeoBot:

Frontend: The web-based user interface will be developed using React.js (v18), a component-based JavaScript library for building interactive user interfaces. The interactive campus map will be implemented using Leaflet.js (v1.9), an open-source JavaScript library for mobile-friendly interactive maps.

Backend: The server-side application will be developed using Node.js (v20 LTS) with the Express.js (v4) framework to handle API requests and route user queries to the appropriate AI processing modules.

Database: Supabase, an open-source backend-as-a-service platform built on PostgreSQL, will be utilized as the primary centralized database for the system. It provides built-in support for RESTful APIs, authentication, and real-time data synchronization, significantly streamlining backend development. Furthermore, rather than relying on an external vector database for the RAG pipeline, the system will utilize the pgvector extension within Supabase. This allows the document embeddings to be stored, indexed, and queried natively within the same PostgreSQL environment as the geospatial campus data and faculty schedule records, resulting in a unified and efficient data architecture.

Machine Learning: The Random Forest classifier will be implemented using Python 3.11 with the scikit-learn (v1.4) library, a widely used machine learning framework that provides robust tools for model training, evaluation, and deployment. The trained model will be served through a lightweight Flask API microservice, which exposes classification endpoints that the Node.js backend can invoke via internal HTTP requests. This microservice architecture ensures separation of concerns between the web application layer and the machine learning inference layer.

Large Language Model and Embeddings: The RAG pipeline will utilize Meta's Llama 3.1 8B as the primary Large Language Model for response generation, accessed through the Groq API for optimized inference speed. Llama 3.1 8B was selected for three reasons: (1) it is an open-source model released under a permissive license, ensuring full transparency and reproducibility of the research; (2) it demonstrates strong instructionfollowing and context-grounding capabilities suitable for retrieval-augmented tasks; and (3) it can be accessed at no cost through free-tier cloud inference providers such as Groq, making it viable for budget-constrained academic research. For document vectorization, the system will employ the all-MiniLM-L6-v2 sentence transformer model, which produces 384-dimensional vector embeddings optimized for semantic similarity search. This model was chosen for its lightweight inference requirements and proven effectiveness in semantic retrieval benchmarks, enabling efficient embedding generation without requiring dedicated GPU infrastructure.

Development Tools: Visual Studio Code will serve as the primary code editor. Git will be used for version control throughout the development process.

Table 1. Software Specification

Software / Technology	**Purpose**	**Feasibility**
React.js (v18)	Frontend user interface	Feasible
Leaflet.js (v1.9)	Interactive campus map	Feasible
Node.js (v20 LTS) + Express.js (v4)	Backend application server	Feasible
Supabase (PostgreSQL + pgvector)	Database and vector store	Feasible
Python 3.11 + scikit-learn (v1.4)	Random Forest classifier	Feasible
Flask API	ML microservice endpoint	Feasible

Llama 3.1 8B (via Groq API)	Large Language Model	Feasible
all-MiniLM-L6-v2	Document embeddings	Feasible
RAGAS Framework	RAG pipeline evaluation	Feasible
Visual Studio Code + Git	Development and version control	Feasible

Table 2. Hardware Specification

Component	Proposed Specification	Feasibility
Processor	Intel Core i5 / i7 (Quad-core)	Feasible
RAM	8.00 GB (Minimum)	Feasible
Storage	256 GB SSD (Minimum)	Feasible
Operating System	Windows 10 / 11 (64-bit) or Linux	Feasible
Internet	Stable broadband connection (for Groq API)	Feasible

3.8 Evaluation Instrument

3.8.1 Technical AI Evaluation (RAGAS Framework)

To objectively evaluate the core research contribution of this study—the Enhanced RAG architecture—the researchers will employ the RAGAS (Retrieval-Augmented

Generation Assessment) framework (Es et al., 2024). RAGAS is a widely adopted automated evaluation framework designed specifically for assessing the quality of RAG pipelines. The evaluation will compare the outputs of the standard RAG pipeline (retrieval + LLM only) against the Enhanced RAG pipeline (retrieval + Random Forest classfication + LLM) using a curated test dataset of representative campus queries. The following four metrics will be computed for each pipeline: (a) Context Precision, which measures the proportion of retrieved document chunks that are actually relevant to the query, thereby evaluating the signal-to-noise ratio of the retrieval step; (b) Context Recall, which measures the degree to which all ground-truth relevant information is successfully retrieved by the pipeline, ensuring comprehensive coverage; (c) Faithfulness, which measures the degree to which the generated response is factually grounded in the retrieved context, directly quantifying the reduction in AI hallucinations; and (d) Answer Relevancy, which measures how directly and completely the generated response addresses the user's original query. The comparative results between the standard RAG and Enhanced RAG across these four metrics will serve as the primary quantitative evidence for the effectiveness of the proposed architectural enhancement.

$$Faithfulness = \frac{|Supported\ Claims|}{|Total\ Claims|}$$

$$Context\ Precision = \frac{\Sigma\ (Precision@k\ \times\ Relevance\_k)}{Total\ Relevant\ Items\ in\ Top - K}$$

Where:

|Supported Claims| = number of claims that can be directly inferred from the retrieved context

|Total Claims| = total number of factual claims made in the generated response

Table 3 summarizes the four RAGAS metrics used in this study. Each metric evaluates a specific dimension of the RAG pipeline's performance, and together they provide a comprehensive assessment of the system's retrieval and generation quality.

Table 3. RAGAS Evaluation Metrics

Metric	Description
Context Precision	Measures the proportion of retrieved context chunks that are actually relevant to answering the query, penalizing irrelevant retrievals.
Context Recall	Measures the retriever's ability to fetch all necessary information by comparing retrieved context against the ground-truth answer.
Faithfulness	Measures whether every factual claim in the generated response can be directly inferred from the retrieved context, detecting hallucinations.
Answer Relevancy	Measures whether the generated response directly and completely addresses the user's original query without including unnecessary information.

Context Recall. This metric measures whether the retrieval component successfully fetched all the necessary information from the knowledge base. It compares the ground-truth answer against the retrieved context to determine what proportion of the expected information was actually retrieved:

$$Context\ Recall\ =\frac{|Ground\ Truth\ Claims\ Supported\ by\ Retrieved\ Context|}{|Total\ Claims\ in\ Ground\ Truth|}$$

Where:

|Supported by Context| = number of ground-truth claims that can be attributed to the retrieved context

|Total Claims in Ground Truth| = total number of factual claims in the ground-truth answer

Answer Relevancy. This metric evaluates whether the generated response directly and completely addresses the user's original query. It works by using the LLM to generate synthetic questions from the response, then computing the mean cosine similarity between each generated question embedding and the original query embedding. A high score indicates the response stays focused on what the user actually asked:

$$Answer\ Relevancy\ =\frac{1}{n}\sum_{i=1}^{n}\ cos(q_i,q)$$

Where:

q_i = embedding of each generated question

q = original user query embedding

n = total number of generated questions

3.8.2 Faculty Functional Validation

To complement the automated RAGAS evaluation, a Functional Validation will be conducted with the 15 selected faculty members and department staff. Each faculty validator will interact with ISU-GeoBot over a defined evaluation period, during which they will query the system regarding their own availability status at various times throughout the day. For each query, the faculty member will record: (a) the system's estimated availability status, (b) their actual status at that time, and (c) whether the classification was correct, partially correct, or incorrect. This structured validation checklist will generate a ground-truth dataset from which the system's Classifcation Accuracy Rate, Precision per status category, and Recall per status category will be computed. This human-verified evaluation provides ecological validity that automated metrics alone cannot capture, as it tests the system's probability estimates against real-world faculty behavior in a live campus environment.

3.9 Statistical Treatment of Data

The data collected from the study will be analyzed using the following approaches corresponding to the two evaluation instruments. For the Technical AI Evaluation, the RAGAS metrics—Context Precision, Context Recall, Faithfulness, and Answer Relevancy—will be computed automatically by the RAGAS framework and reported as comparative scores (0.0 to 1.0) for both the standard RAG and Enhanced RAG pipelines. The improvement will be expressed as the percentage difference in mean scores across the curated test queries, accompanied by visualizations (bar charts and radar plots) to illustrate the comparative performance. For the Faculty Functional Validation, the following metrics will be computed from the ground-truth validation checklists:

Classification Accuracy Rate. The overall classification accuracy will be computed as the ratio of correct classification to total classification verified by faculty validators: Accuracy = (Number of Correct Classification / Total Classification Verified) × 100%.

This metric provides a direct, human-verified measure of the system's ability to correctly predict faculty availability status.

$$Accuracy = \frac{Number\ of\ Correct\ Classification}{Total\ Predictions\ Verified} \times 100\%$$

Where:

Correct Classification = number of classification matching faculty-confirmed actual status

Total Classification = total number of classification verified by faculty validators

Per-Category Precision and Recall. For each availability status category ("Available for Consultation," "Currently in a Lecture," "Unavailable"), precision and recall will be calculated to determine the system's accuracy at the category level. Precision measures the proportion of classifications for a given category that were correct, while recall measures the proportion of actual instances of a category that were correctly identified. These metrics will be presented in a confusion matrix format to provide a comprehensive view of probability performance across all status categories.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Where:

Precision = proportion of classifications that were correct

Recall = proportion of actual instances correctly identified

3.10 Ethical Considerations

This study will adhere to ethical standards in both the development of the system and the conduct of the evaluation. The following ethical considerations will be observed:

Informed Consent. All faculty validators will be informed about the purpose of the study, the nature of their participation, and their right to withdraw at any time without penalty. Written informed consent will be obtained from each faculty validator prior to the evaluation.

Data Privacy. In compliance with Republic Act No. 10173 (Data Privacy Act of 2012), no personally identifiable information of faculty members will be stored, transmitted, or displayed by the system. The status masking protocol described in Section 3.5.4 ensures that the system outputs only generalized availability statuses rather than exact physical locations. Survey responses will be collected anonymously to protect respondent identity.

Voluntary Participation. Faculty participation in the functional validation will be entirely voluntary. No professional incentive or penalty will be associated with participation or non-participation in the study.

REFERENCES

Alam, S., Sari, R. M., Alfian, G., & Farooq, U. (2024). Room occupancy detection based on random forest with timestamp features and ANOVA feature selection method. Journal of Computing Science and Engineering, 18(1), 10–18. https://doi.org/10.5626/JCSE.2024.18.1.10

Brown, A., Roman, M., & Devereux, B. (2025). A systematic literature review of retrieval-augmented generation: Techniques, metrics, and challenges. Big Data and Cognitive Computing, 9(12), Article 320. https://doi.org/10.3390/bdcc9120320

Budianto, A. E., & Othman, M. F. I. (2025). An integrated framework for improving online counseling system quality: A case study using HOT-FIT, ISO/IEC 25010, and SPBE models. International Journal of Research and Innovation in Social Science, 9(23), 100–124. https://doi.org/10.47772/IJRISS.2025.9230009

Cheong, P. H., & Nyaupane, P. (2022). Smart campus communication, Internet of Things, and data governance: Understanding student tensions and imaginaries. Big Data & Society, 9(1), 1–14. https://doi.org/10.1177/20539517221092656

Creswell, J. W., & Creswell, J. D. (2018). Research design: Qualitative, quantitative, and mixed methods approaches (5th ed.). SAGE Publications.

Dey, A. K. (2001). Understanding and using context. Personal and Ubiquitous Computing, 5(1), 4–7. https://doi.org/10.1007/s007790170019

Dietterich, T. G. (2000). Ensemble methods in machine learning. In J. Kittler & F. Roli (Eds.), Multiple classifier systems: First international workshop, MCS 2000 (pp. 1–15). Springer. https://doi.org/10.1007/3-540-45014-9_1

Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2024). RAGAS: Automated evaluation of retrieval augmented generation. In Proceedings of the 18th Conference of the European Chapter of the Association for Computational

Linguistics: System Demonstrations (pp. 150–163). Association for Computational Linguistics. https://aclanthology.org/2024.eacl-demo.16/

Faulkner, L. (2003). Beyond the five-user assumption: Benefits of increased sample sizes in usability testing. Behavior Research Methods, Instruments, & Computers, 35(3), 379–383. https://doi.org/10.3758/BF03195514

Haggag, M., Oulefki, A., Amira, A., & Foufou, S. (2025). Integrating advanced technologies for sustainable smart campus development: A comprehensive survey of recent studies. Advanced Engineering Informatics, 65, Article 103412. https://doi.org/10.1016/j.aei.2025.103412

Hoang, V. N., Dai, N. N., Kim, T. N. T., Dinh, T. H., Chi, T. P., Minh, T. T., & Xuan, V. T. (2025). Enabling smart campus indoor spaces through spatial modeling with the IMDF platform. CTU Journal of Innovation and Sustainable Development, 17(Special Issue: ISDS), 88–96. https://doi.org/10.22144/ctujoisd.2025.056

Kanapathy, P., Mustafa, N., Yusop, N., Ibrahim, Y., Moketar, N. A., & Hakimi, H. (2026). Enhancing attendance authenticity in higher education using a QR code and GPS-based smart attendance system. International Journal of Research and Innovation in Social Science, 10(1), 7855–7867.

Kaur, H., Malhotra, C., Pandey, S., & Sahu, R. (2025). Smart campus development: Integrating healthcare, engineering, IT, business, and legal frameworks for future education. Lex Localis – Journal of Local Self-Government, 23(S4). https://doi.org/10.4335/23.s4.801162

Lang, G., & Gürpinar, T. (2025). AI-powered learning support: A study of retrieval-augmented generation (RAG) chatbot effectiveness in an online course. Information Systems Education Journal, 23(2), 4–16. https://files.eric.ed.gov/fulltext/EJ1467995.pdf

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. Advances in Neural Information Processing Systems, 33, 9459–9474. https://doi.org/10.48550/arXiv.2005.11401

Mahdhin, D., Mohammed, L., Maaz, O. M., & Alwarafy, A. (2025). Enhancing the precision and interpretability of retrieval-augmented generation (RAG) in legal technology: A survey. IEEE Access, 13, 1–20. https://doi.org/10.1109/ACCESS.2025.3550145

Mazzei, S., Zambotto, L., Tealdo, G., Macagno, A., & Palmero Aprosio, A. (2025). Uni-Mate: A retrieval-augmented generation system to provide high school students with accurate academic guidance. In Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025) (pp. 699–709). Association for Computational Linguistics. https://aclanthology.org/2025.clicit-1.67.pdf

Nayakwadi, V., Supekar, V., Takrani, B., & Thakare, A. (2025). Smart campus navigation system: Enhancing wayfinding using AI and IoT integration. International Journal for Research in Applied Science and Engineering Technology, 13(12). https://doi.org/10.22214/ijraset.2025.75716

Nielsen, J. (1994). Usability engineering. Morgan Kaufmann Publishers.

Richey, R. C., & Klein, J. D. (2014). Design and development research: Methods, strategies, and issues. Routledge.

Sommerville, I. (2016). Software engineering (10th ed.). Pearson.

Swacha, J., & Gracel, M. (2025). Retrieval-augmented generation (RAG) chatbots for education: A survey of applications. ResearchGate. https://doi.org/10.13140/RG.2.2.22086.52804

Zhang, L., Elbertsen, M., Kok, H., & Salimi, N. (2025). Designing the future smart campus: Integrating key elements to enhance user experience. Journal of Science and Technology Policy Management, 16(10), 117–137. https://doi.org/10.1108/JSTPM-03-2024-0054